

The sampling frame

Fictional campus printers:

North: N1, N2, N3, N4, N5, N6

South: S1, S2, S3, S4, S5, S6

East: E1, E2, E3, E4, E5, E6

West: W1, W2, W3, W4, W5, W6

How could we select printers to inspect?

A simple random sample

Use the 24-printer list from the first slide. Label the printers 01-24 in the displayed order.

How can we randomly select 6 distinct printers so every group of 6 is equally likely?

A systematic sample

Use the same 24-printer list. Choose a random starting position from 1 through 4, then take every fourth printer.

Which 6 printers enter our sample?

Stratifying by building

North, South, East, and West each contain 6 printers.

How can we randomly select one-third of the printers in each building?

A cluster sample

Treat each of the four buildings as a cluster.

How can we randomly select 2 buildings and inspect every printer in them?

Same groups, different method

Plan A selects 2 printers from each building.
Plan B selects every printer in 2 randomly chosen buildings.

How do the two selection procedures differ?

Unequal strata

A second campus has 12 printers in North, 8 in South, and 4 in East.

For a proportional stratified sample of 6, how many should come from each building?

Ordering matters

A list repeats this sequence: color, black-and-white, black-and-white, black-and-white.

What could happen if we sample every fourth printer?

A reproducible procedure

Another group must recreate our selection process using a fresh random draw.

What details must our sampling description include?

Choosing a method

Travel between buildings is costly, but printers within one building may be very similar.

Which sampling method would you choose, and what tradeoff would you accept?